

“Bodyreath”: An Interactive VR Installation to Experience and Understand People with Autism

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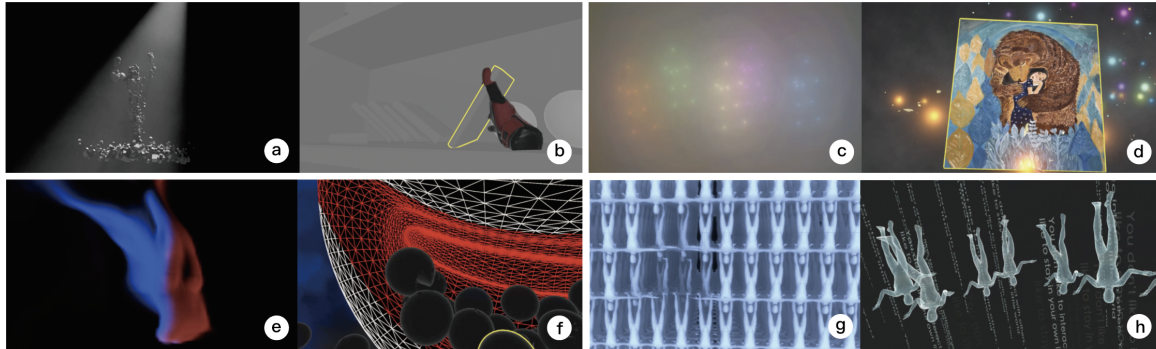


Figure 1: “Stereotyped Behavior”: (a) The virtual avatar. (b) The VR scenario, “Children from the Stars”: (c) The virtual avatar. (d) The VR scenario, “Emotional Instability”: (e) The virtual avatar. (f) The VR scenario, “Invisibility”: (g) The virtual avatar. (h) The VR scenario

ABSTRACT

“Bodyreath” is an interactive virtual reality (VR) installation for understanding people with autism, which is divided into two parts, “external” and “internal”. The “external” is the external projection, which is the artistic effect of four virtual avatars based on the characters and behaviors of autistic people. The “internal” is the interactive VR scenarios, allowing the audience to gain insight into the characteristics of the autistic people represented in each feature: “Stereotyped Behavior”, “Children from the Stars”, “Emotional Instability”, and “Invisibility”. Our VR installation provides an interactive narrative through body interaction and role perception, allowing the audience to transform their identity into an autistic person and perceive it from their perspective. It helps the audience to appreciate the innocence of autistic people in a different way and to understand them artistically.

CCS CONCEPTS

• **Human-centered computing** → Virtual reality; • **Applied computing** → Media arts.

KEYWORDS

VR installation, body interaction, people with autism, motion capture

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1 INTRODUCTION

The number of autistic people around the world is increasing year by year, and they are generally poorly understood by the public. VR is enabling to realize a multi-dimensional interactive experience in digital arts [Gonzalez-Franco et al., 2019][Adjorlu and Serafin, 2019]. To make the public understand and concern more about this community, we designed “Bodyreath”, an interactive VR installation. We have spent nearly a year of research work in a public nonprofit organization to understand and analyze the behavioral characteristics of autistic people. Motion capture devices were used to capture their dance movements and visualize them in real time [Jürgens et al., 2020][Cisneros et al., 2019] to create four different artistic effects. Based on this, we designed four VR scenarios that represent the characteristics of autistic people, which use body interaction and spatial audio technology to enable the audience to understand people with autism from different perspectives. The demonstration of this installation was conducted in June 2021 and received positive feedback.

2 IMPLEMENTATION

“Bodyreath” is mainly divided into two parts, namely the external projection and the internal VR scenarios. Firstly, we design the avatars according to four characteristics of people with autism. We use the motion capture device of NoitomTM to capture their dance movements of people with autism in real time and use the visualization tool “Notch” to create the artistic effect of the avatars. The performance of the avatars were projected onto a customized four-sided projection screen, and the audience can enjoy their dance from outside the installation. When the audience enter the interior of the installation, wearing VR headsets, handles, and wearable trackers, they can select and enter four virtual scenarios with similar characteristics to the external projection (Figure 1).

3 VIRTUAL SCENARIOS

3.1 Stereotyped Behavior

The first scenario represents the characteristic of stereotyped behavior. When the audience enter this space, any object can be grabbed unrestrictedly, but for every object grabbed, it will turn into a ball. Some objects will turn into a ball dropping on the ground due to gravity, like books, pens, pillows, etc., while the fruit in the potted plant will be suspended in the air after picking and the paper airplane float in the air without gravity. When picking up a paintbrush to touch the drawing board, a circle will be drawn on it. This scenario wants to let the audience feel the monotonous and stereotyped cognitive behavior of people with autism.

3.2 Children from the Stars

People with autism are called “children from the stars” as they have distinctive features, namely unique individual aesthetics and creativity. In this scenario, the audience enter into a colorful world full of stars and can roam around the space. When their hands touch the yellow stars, the paintings created by the autistic people will appear. There are a total of 42 digital paintings in the scene (the World Autism Awareness Day is on April 2nd, hence 42 hidden paintings are placed in this scene), all of which are 3D reconstructions based on real paintings. There are planets in different positions above the starry sky to enable point-to-point transportation.

3.3 Emotional Instability

The third scenario focuses on the emotional instability of autistic people and the difficulty of conveying other’s emotions to them. When the audience enter the scenario, they will see many black expression balls. Through body interaction, the audience can pick up these balls and throw them towards the animated “red and blue” expression area, or kick them over, and the surrounding environment will spew red or blue smoke according to collision detection, presenting that normal emotions are not colored in the world of people with autism, but colorless, and when these colorless emotions are conveyed to them, the only two responses they can make, good or bad.

3.4 Invisibility

The fourth scenario concentrates on “invisibility”, where people with autism are invisible or ignored to most people. In this scenario,

we use distance detection and spatial audio technology for interaction. When the audience approaches the avatar, the avatar will make an audio response related to autism and the virtual environment will change based on textual content. The audio feedback is based on a preliminary questionnaire on the perception of people with autism. Therefore, the audience could have a comprehensive understanding of the public’s perception of people with autism and make their own judgments.

4 SUMMARY

This installation was experienced by 50 people. The feedback is significant different between adults and children. Children spent a longer time to repeat the stereotyped behavior, reflecting that the interests of people with autism are more consistent with those of children. In the third scene, some participants showed fear, which is the same as the first connection with autistic people with emotional instability. “Bodyreath” allows the public to appreciate the innocence of people with autism in multiple dimensions, to understand and experience their cognitive style, and to pay more attention to this community.

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